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High-energy tunable ultraviolet pulses generated by optical leaky wave in filamentation



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Ultraviolet pulses could open up new opportunities for the study of strong-field physics and ultrafast science. However, the existing methods for generating ultraviolet pulses face difficulties in fulfilling the twofold requirements of high energy and wavelength tunability simultaneously. Here, we theoretically demonstrate the generation of high-energy and wavelength tunable ultraviolet pulses in preformed gas-plasma channels via the leaky wave emission. The output ultraviolet pulse has a tunable wavelength ranging from 91 nm to 430 nm, and an energy level up to sub-mJ. Such a high-energy tunable ultraviolet light source may provide promising opportunities for characterization of ultrafast phenomena, and also an important driving source for the generation of high-energy attosecond pulses.

Ultraviolet (UV) pulses have gained increasing popularity in the last few decades due to their versatile applications in biology¹, material fabrication², and spectroscopy³. With the rapid development of laser technology, UV pulses in femtosecond scale have been generated and widely applied⁴. Within the entire UV range, the UV pulses with wavelengths from 100 to 400 nm remain highly required because they can be resonantly absorbed by most substances and can serve as unique light sources to study the ultrafast dynamics of atoms and molecules, for instance, measuring absolute transition frequency of atoms⁵, revealing structure and dynamics of biomolecules⁶, and exciting neutral chiral molecules⁷. Moreover, when the UV pulse with high energy and tunable wavelength is used as a single driving source or one of the two-color fields, it enables the generation of high-order harmonics with high conversion efficiency^{8,9}, which will be beneficial to applications that require high photon flux, such as attosecond spectroscopy¹⁰ and extreme-UV lithography¹¹.

At present, UV pulses in the spectral region of 100 to 400 nm are mainly generated from low-order harmonics¹² and four-wave mixing¹³. Nevertheless, the wavelength of the generated UV pulse is often limited by the available wavelength of the pump pulse. To overcome this problem, UV pulses with continuously tunable wavelength can be generated from resonant dispersive wave in gas-filled hollow-core fibers by varying the gas pressure^{14,15}. In this case, the energy of UV pulses has reached a few tens of μJ , but it is difficult to be further enhanced due to the limited length and diameter of the fiber and the available energy of the few-cycle pump pulses¹⁶. Although the removal of the fiber confinement could break the restriction on energy enhancement, it also leads to the lack of negative dispersion provided by fiber waveguides. Correspondingly, UV pulses cannot be obtained because the propagation of the near-infrared femtosecond laser pulse in uniform gas results in supercontinuum emission with cut-off wavelength of ~ 400 nm.

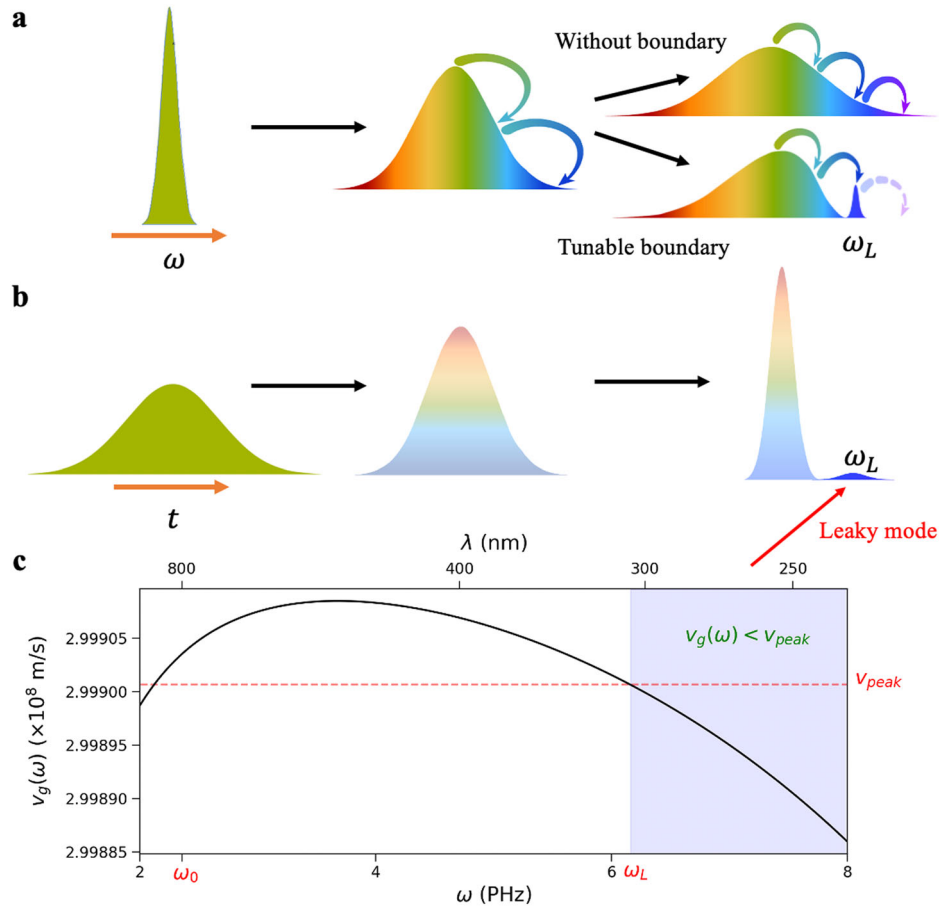
In this article, we propose to generate wavelength-tunable UV pulses from the supercontinuum emission of the near-infrared femtosecond pulse by introducing a preformed plasma channel with the adjusted negative dispersion into air and noble gases. The output UV pulse has a tunable wavelength ranging from 91 to 430 nm and an energy level up to sub-mJ, suggesting that this approach holds great potential in the generation of widely tunable UV pulses with rather high energy.

Results and discussion

Concept

Our basic idea is illustrated in Fig. 1. During filamentation in air, spectral broadening induced by self-phase modulation can be viewed as a cascaded process. The spectral energy is transferred from the central frequency to high frequency components, and then higher. This process could lead to supercontinuum generation with blue-side cut-off wavelength ~ 400 nm, which is the common situation for air filamentation. The spectrum in UV region cannot be obtained because air is normally dispersive, and the high-frequency components move far behind the pulse peak gradually, with their intensity too low to participate in further frequency shift. To overcome this limitation, we employ a preformed uniform plasma channel, which can introduce negative dispersion into air. Due to this negative dispersion, the high-frequency components can be constrained near the peak of the pulse. They can undergo further frequency shift to the UV region. As shown in Fig. 1a, the cascaded frequency shift combined with the negative dispersion could result in the super-broad spectrum, which covers the UV region. To obtain an isolated UV spectral hump, we need a frequency boundary ω_L to terminate the cascaded process. For the components with frequency higher than ω_L , they fall behind the primary pulse and will not undergo further frequency shift due to the relatively low intensity. Therefore, the spectral

Fig. 1 | Schematic illustration of leaky wave generation. **a** The spectral broadening process of filamentation when there is no spectral boundary (top) and a spectral boundary (bottom). **b** The pulse self-compression and leaky wave generation in the tail of the pulse. **c** The group velocity v_g as a function of frequency ω when the plasma density is $\rho_0 = 1.5 \times 10^{17} \text{ cm}^{-3}$. The red dashed line denotes the velocity of pulse peak v_{peak} . The colored area represents the leaky mode, with $v_g(\omega) < v_{\text{peak}}$.



energy is deposited in this region, as shown in Fig. 1. We term this process temporal leaky wave emission, because it is similar to the spatial leaky wave in the waveguide¹⁷.

We will show the frequency boundary ω_L of the cascaded broadening, which corresponds to the peak frequency of the leaky UV spectrum, could be tuned by adjusting the density of the preformed plasma channel. To calculate ω_L , we need to find the spectral components which fall behind the primary pulse. Here we use a preformed air-plasma channel with plasma density ρ_0 to provide tunable dispersion condition, and the dispersion relation is given by

$$k(\omega) = \frac{n_a(\omega)\omega}{c} - \frac{\rho_0\omega}{2c\rho_c}, \quad (1)$$

where $n_a(\omega)$ is the refractive index of air, and $\rho_c = \epsilon_0 m_e \omega^2 / e^2$ is the frequency-dependent critical plasma density that accounts for plasma dispersion. For the velocity of the primary pulse, we use the group velocity of the pulse peak with central frequency ω_0 and consider the self-steepening effect. It is written as

$$v_{\text{peak}} = \frac{1}{\partial_\omega k(\omega_0) + n_2 I_p / c}, \quad (2)$$

where n_2 is the nonlinear refractive index coefficient and I_p is the peak intensity of the pulse. The group velocity of spectral components falling behind the primary pulse is given by $v_g(\omega) < v_{\text{peak}}$, and the critical frequency ω_L satisfies

$$v_g(\omega_L) = v_{\text{peak}}. \quad (3)$$

From the above analysis, we can see that the peak frequency of the UV spectrum ω_L could be tuned by adjusting the plasma density ρ_0 .

To verify that the dispersion induced by the preformed low-density plasma channel can indeed support the generation of UV pulses via temporal leaky-wave emission, we calculate the group velocity of different spectral components for the plasma density $\rho_0 = 1.5 \times 10^{17} \text{ cm}^{-3}$, $n_2 = 0.96 \times 10^{-19} \text{ cm}^2/\text{W}$ and $I_p = 100 \text{ TW}/\text{cm}^2$, as shown in Fig. 1c. For the blueshift components with frequency smaller than ω_L , their group velocities are larger than the pulse peak. Although they are generated at the pulse trailing edge by self-phase modulation, they can be confined within the primary pulse. For spectral components with frequency larger than ω_L , their group velocities are lower than the pulse peak. Therefore, this part of the spectrum will lag behind the primary pulse. This evolution of group velocity as a function of the frequency supports our scheme of the temporal leaky-wave emission. For the plasma density $\rho_0 = 1.5 \times 10^{17} \text{ cm}^{-3}$, the corresponding wavelength of the leaky-wave emission is 320 nm, located in the UV region.

To confirm our model, we numerically solve the UPPE (see “Methods”) to study the propagation of femtosecond laser pulses in gas-plasma channels. The input 800 nm pulse has a duration of 30 fs, a beam waist of 2 mm and a peak power of $P_{\text{in}} = 5P_{\text{cr}}$, where P_{cr} is the critical power for self-focusing¹⁸. For air, $P_{\text{cr}} = 10 \text{ GW}$ and the input pulse energy is 1.5 mJ. The beam is focused by a lens with $f = 1 \text{ m}$.

Leaky wave generation

Firstly, we investigate the pulse dynamics in air when the density of preformed plasma channel $\rho_0 = 1.5 \times 10^{17} \text{ cm}^{-3}$. Filamentation can be divided into three stages. At $z = 1 \text{ m}$, we can see supercontinuum generation (Fig. 2a) as well as pulse compression (Fig. 2b). Pulse-splitting also occurs at this stage (Fig. 2b), and it can be seen that the supercontinuum is mainly contributed

by the rear sub-pulse. Since the supercontinuum has reached the critical wavelength of leaky wave, at $z = 1.1$ m there arises a high-frequency lobe in the spectrogram, which corresponds to the newly generated sub-pulse in Fig. 2b and the spectral hump at 283 nm in Fig. 2c. Then the leaky wave is continuously generated and lagged behind, which makes the time span and the spectral intensity of leaky wave increase. As the spectral components between ω_0 and ω_L are consumed, we can see the formation of isolated spectral hump. The simulation results confirm our scheme, except for a slight difference in the peak wavelength of the UV hump. The reason for this difference will be discussed below.

To study the temporal properties of the output UV pulse, the spectral components between 200 and 350 nm are filtered, and inverse Fourier transform is made to obtain the temporal profile. Figure 3 shows the temporal and spectral profiles of the output UV pulse (on-axis) at different propagation distances. At the initial stage of leaky wave generation ($z = 1.05$ m), the intense part of the spectrum has a nearly flat phase, and the corresponding pulse duration is about 2.3 fs, close to that of the transform limit pulse. However, as the leaky wave is continuously generated, the UV pulse becomes positively chirped and the pulse is gradually stretched because the leaky wave locates at the positive dispersion region, in accordance with Fig. 2a. For applications that require ultrashort UV pulses, the pre-plasma channel may be gradually terminated after the focus via a negative gas pressure gradient in a gas cell¹⁹. In this case, there should be a compromise between pulse duration and energy. Since the positive dispersion is merely contributed by the gas itself, we may also use gases with weaker dispersion in the UV range, such as helium, to reduce the pulse duration, as we will show later.

Frequency tuning

For different preformed plasma densities ρ_0 , the output spectra are shown in Fig. 4a. When $\rho_0 < 0.3 \times 10^{17} \text{ cm}^{-3}$, the spectrum is broadened without isolated UV spectral hump. As the density of the pre-plasma increases, the stronger negative dispersion leads to pulse self-compression, resulting in a higher peak intensity and a shorter pulse duration, as shown in Fig. 4b. This gives rise to a much steeper intensity slope $\partial_t I$, thus a much broader spectrum, which covers the critical wavelength of leaky wave $\lambda_L = 2\pi c/\omega_L$ and an isolated UV spectral hump is formed. The central wavelength of the UV spectral hump decreases as ρ_0 increases, resulting in a wavelength tuning

range between 250 and 430 nm. The simulation results accord well with the theoretically predicted value of λ_L from Eq. (3), denoted by the red dashed line in Fig. 4a. The deviation between our model and simulation mainly comes from the post-frequency-shift process of leaky wave: the spectral component with wavelength of λ_L generated near the pulse peak undergoes further frequency shift as it moves toward the pulse tail. The lower bound of wavelength tuning in our scheme is about 250 nm, which is limited by two factors. For one thing, the conversion efficiency decreases with the output wavelength. For another, the clamping intensity limits the spectral width of the supercontinuum, while to generate leaky wave, the supercontinuum has to cover the critical wavelength. The above restriction may be broken by using a pump beam with shorter central wavelength. The use of gases with weaker dispersion may also be a good choice, because the weaker dispersion could extend the constrained distance of the broadened spectrum with the primary pulse, leading to the enhancement of post-frequency-shift process mentioned above.

To generate a long uniform pre-plasma channel with variable density in the 10^{17} cm^{-3} level, collisional ionization based scheme is a promising candidate. Here we give a proof-of-principle simulation to generate the pre-plasma channel required by our scheme with high-energy picosecond CO_2 laser pulses, which have been used to generate megafilament in air²⁰. The input beam is unfocused, with a fourth-order super-gaussian profile and a diameter ($1/e^2$) about 3.2 cm. The pulse energy is changed between 1.84 J and 1.92 J, and the pulse duration is 2 ps, resulting in peak intensity $\sim 0.2 \text{ TW/cm}^2$. The central wavelength is taken to be 10 μm , and the non-linear coefficient $n_2 = 5 \times 10^{-19} \text{ cm}^2/\text{W}^{20}$. The ionization rate is composed of three parts: the PPT rate that considers multi-photon and tunnel ionization, the excitation induced dephasing (EID) contribution and the collisional ionization (see “Methods”). The EID effect could induce considerably large ionization rate compared to the PPT rate when the laser intensity is relatively low²⁰. As a consequence, seed electrons are generated, and later collisional ionization further increases the plasma density to a high level. As shown in Fig. 4c, the maximum plasma density could be tuned by changing the pulse energy. More importantly, for a given pulse energy, the plasma density only varies slightly in a meter scale longitudinally and in a centimeter scale transversely. For single-filament usage, a pre-plasma channel with mm-scale diameter should be enough, because the diameter of infrared laser

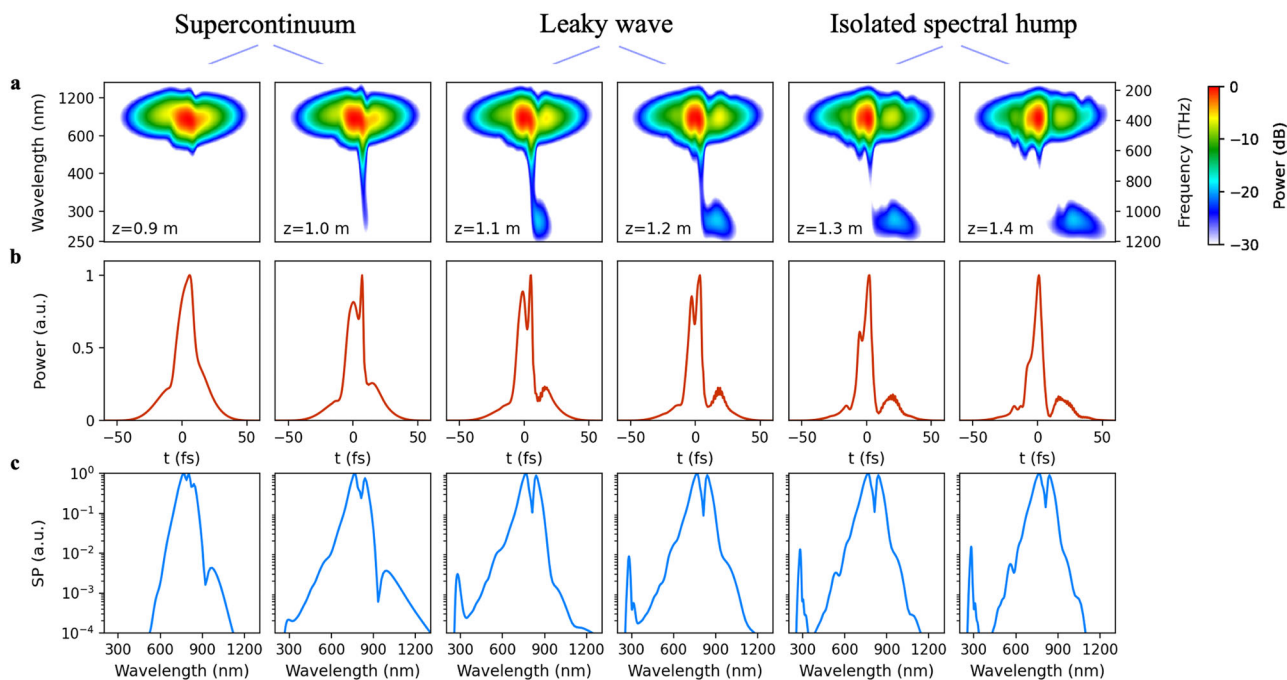


Fig. 2 | Schematic of leaky wave generation in a preformed air-plasma channel. **a** Spectrogram, **b** pulse profile, and **c** spectral power (SP) distribution at different positions. The density of the air-plasma channel is $\rho_0 = 1.5 \times 10^{17} \text{ cm}^{-3}$.

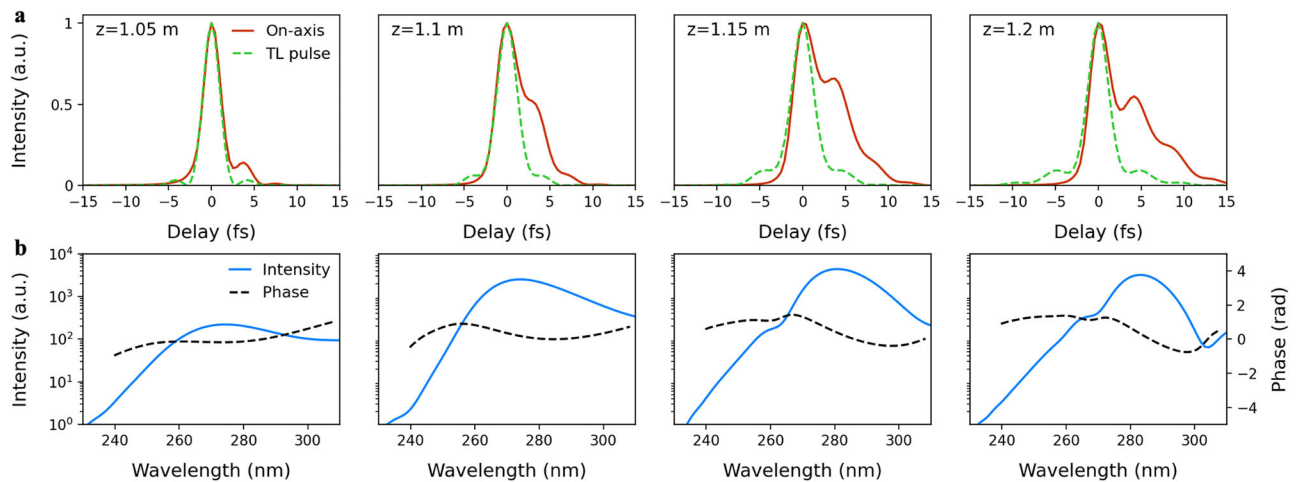


Fig. 3 | UV pulse profile. **a** The on-axis and the corresponding transform limit (TL) pulse profile of the UV pulses. **b** The on-axis spectral intensity and phase of the UV pulses. Spectral components between 200 and 350 nm are filtered to obtain the UV pulses. The density of the air-plasma channel is $\rho_0 = 1.5 \times 10^{17} \text{ cm}^{-3}$.

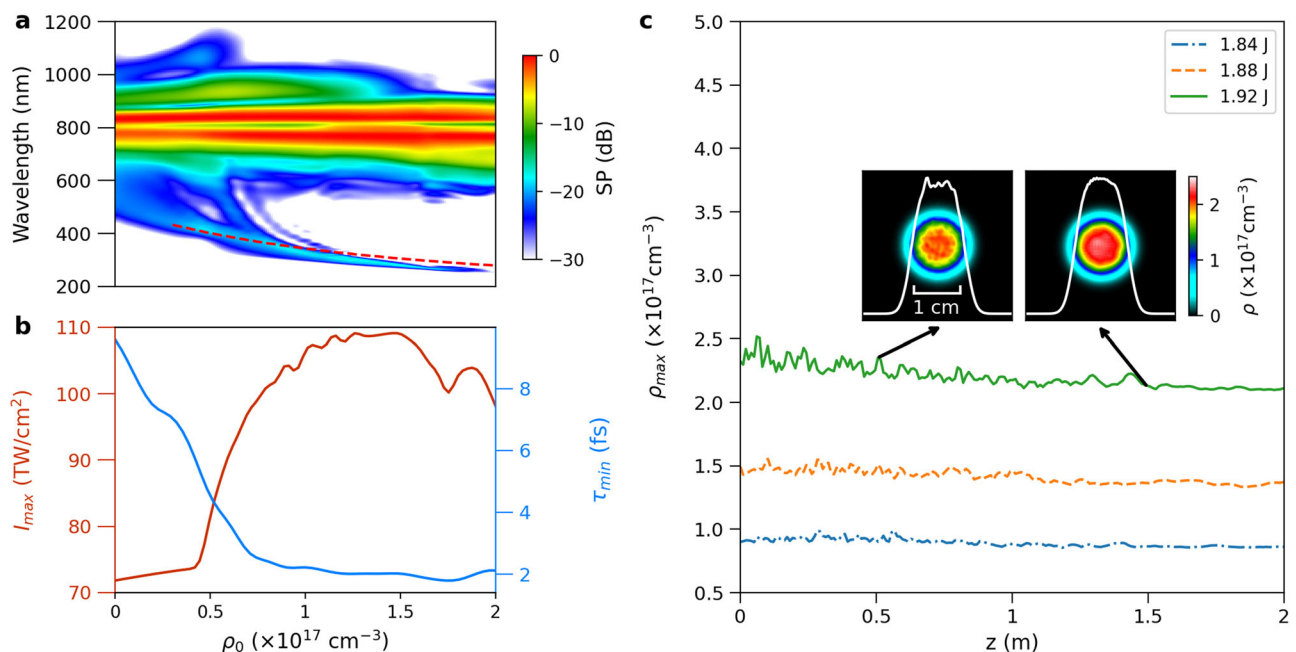


Fig. 4 | Frequency tuning via changing density of air-plasma channels. **a** Output spectral power (SP) distribution as ρ_0 changes from 0 to $2 \times 10^{17} \text{ cm}^{-3}$. The red dashed line denotes the central wavelength of leaky wave predicted by Eq. (3). **b** Maximum intensity (red) and minimum on-axis pulse duration (blue) during

propagation for different values of ρ_0 . **c** Evolution of the maximum plasma density generated by CO₂ lasers with different energy. The inset shows the transverse plasma density distribution at $z = 0.5$ m and $z = 1.5$ m for 1.92 J pulse energy.

filament is typically a few hundreds of μm . In this case, the required energy of the picosecond CO₂ laser pulse may be reduced to tens of mJ, which makes this scheme more feasible. In future experiments, a picosecond CO₂ laser pulse collinearly propagates in front of an infrared femtosecond pulse, with a time delay much shorter than the lifetime of plasma. The high-energy picosecond CO₂ pulse ionizes the gas and generates a long uniform plasma channel, and the subsequent femtosecond pulse generates the UV pulse with this gas-plasma channel. Note that the repetition rate of the picosecond and femtosecond pulses should be kept low, ideally around 10 Hz, because for a high repetition rate the thermal diffusion of the gas after the plasma recombination can introduce undesirable instability into the system.

Energy scaling

As an important criteria of UV light source, the energy of UV pulse is typically a few μJ using photonic crystal fibers and hollow-core capillary

fibers. The fiber system, though providing a simple and flexible platform to generate tunable UV pulse source, also puts an intrinsic limitation on the available energy. Our air-plasma channel scheme could be a promising candidate for obtaining UV pulses with rather high energy, without the concern of damage to the media. To explore the energy scaling rules of our scheme, we first investigate the influence of input pulse peak power on the output spectra. We fix ρ_0 at $1.5 \times 10^{17} \text{ cm}^{-3}$ and gradually increase the input power from P_{cr} to $9P_{\text{cr}}$. The output spectra evolution is shown in Fig. 5a. Leaky wave emission with a wavelength of 280 nm is generated when the input peak power $P_{\text{in}} \approx 1.5P_{\text{cr}}$. When increasing the input power, the leaky wavelength first becomes longer (about 10 nm), because the minimum pulse width is slightly shorter, and the post-frequency-shift effect mentioned before is weaker. For $P_{\text{in}} > 5P_{\text{cr}}$, the minimum pulse width as well as the leaky wavelength is almost constant, which ensures the stability of our scheme. As the incident

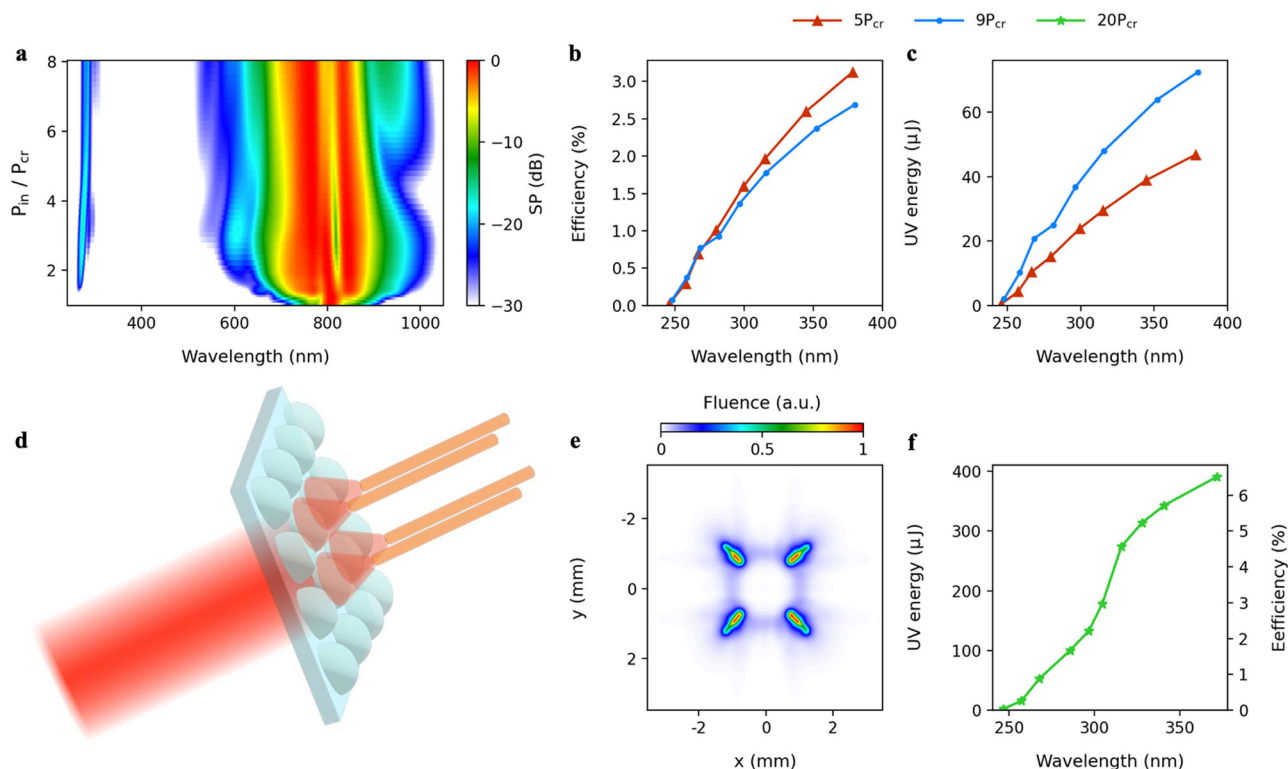


Fig. 5 | Energy scaling proposals of the UV radiation. **a** Output spectral power (SP) distribution as varying the input power. **b** The conversion efficiency and **c** the energy of the output UV pulses for different input power and output wavelength.

d Schematic illustration of regularization of multiple filaments with a microlens array. **e** The fluence distribution and **f** the conversion efficiency and output UV pulse energy when using a microlens array.

power increases, the UV spectral hump becomes more intense with an increased energy and a slightly decreased conversion efficiency. For $P_{in} = 5P_{cr}$ and $9P_{cr}$, the UV spectrum has a central wavelength of 283 and 285 nm, an energy of 13.3 and 22.4 μJ , and a conversion efficiency of 0.88% and 0.83%, respectively. We also calculate the energy and conversion efficiency of the UV spectrum with different central wavelength by adjusting pre-plasma density. As expected, the conversion efficiency is lower for shorter wavelength due to the cascaded feature of spectral broadening, as shown in Fig. 5b. For wavelength shorter than 280 nm, the conversion efficiency is lower than 1%. However, since the conversion efficiency only decreases slightly when the input peak power is increased from $5P_{cr}$ to $9P_{cr}$ (less than 0.5%), energy scaling of UV pulses could be directly achieved by increasing the input power, as indicated by Fig. 5c. In this case, the output energy of UV pulses could be more than 40 μJ for wavelength longer than 300 nm.

If further increasing the input power, the influence of multiple filaments should be considered. Here we propose that a microlens array (MLA) may be used to obtain regularly distributed filaments, as demonstrated in Fig. 5d. Since the supercontinuum emitted from each single filament can be coherently combined²¹, using a MLA to organize the filaments generated by high-energy pulse could be a promising solution to mJ level UV pulse. As a proof-of-principle simulation, we use an incident pulse with peak power $P_{in} = 20P_{cr}$, focused by a MLA (the focal length and size of each square lenslet are 1 m and 2 mm, respectively). Four filaments are formed (Fig. 5e), and the output UV pulse energy reaches sub-mJ level, as shown in Fig. 5f. Note that when using MLA, the conversion efficiency is nearly doubled, about 6% for a 350 nm pulse. This is related to the interaction of the energy background of filaments: the energy that is defocused by one filament could be reused by other filaments. It can also be seen that it is necessary to provide a uniform plasma channel with a centimeter-diameter for the generation of high-energy UV pulses. Such a plasma channel may be generated by filamentation of high-energy picosecond laser pulses²⁰, as discussed before.

UV pulse generation in noble gases

For noble gases, the magnitudes of nonlinear refractive index coefficient, dispersion, and ionization potential are different from air. The use of noble gases may provide a wider tuning range for the generation of UV pulses. To this end, we tested our scheme in gas cells filled with noble gases. The input power is kept at $P_{in} = 5P_{cr}$, where P_{cr} changes with gas species and pressure. Here we present the simulation results of two typical gases, argon and helium. At 1 atm pressure, the critical power is 6 GW for argon and 267 GW for helium, corresponding to input pulse energy of 0.9 mJ and 40 mJ, respectively.

For argon, the dispersion is weaker than air in the UV region. According to the leaky wave emission model, the weaker positive dispersion of gas will lead to a shorter wavelength of the UV hump. As can be seen from Fig. 6a, the wavelength of the isolated UV hump generated in argon at 1 atm pressure covers from 125 to 200 nm (above 200 nm the spectrum is continuous), which is shorter than that in air. In addition, the weaker dispersion also gives rise to a steeper pulse trailing edge and extends the constrained distance of the broadened spectrum with the primary pulse, so that the spectrum could cover the leaky wavelength and the conversion efficiency of the UV spectrum is enhanced. As can be seen from Fig. 6c, for the UV hump in this spectral region (125–200 nm), the conversion efficiency is several times higher than that in air, and the corresponding output energy of the UV pulse could reach several tens of μJ . The temporal duration evolution of the UV pulse is similar to the case in air. For instance, the minimal duration of the UV pulse is 3.5 fs when the leaky wave with wavelength of 168 nm is generated at the initial stage. With the increase of the propagation distance, the pulse is stretched due to the positive dispersion though the dispersion of argon is weaker. To obtain a shorter UV pulse with high energy, we need to use noble gas with much weaker dispersion.

To further increase the energy of the UV pulse generated in argon, we decrease the gas pressure to 0.5 atm. Because the critical power for self-focusing scales as $1/p$, the input energy is 1.8 mJ. For the wavelength above 140 nm, although the conversion efficiency is lower (Fig. 6c), the UV energy remains higher due to the increased input power, reaching around 100 μJ

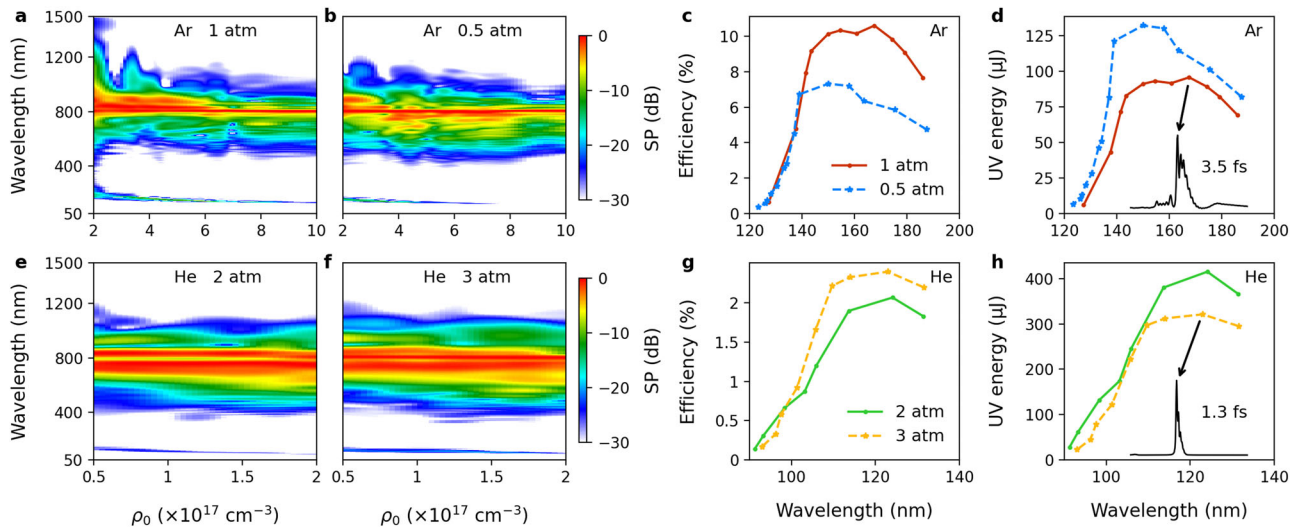


Fig. 6 | Influence of gas species and pressure. Leaky wave is generated in **a–d** argon and **e–h** helium at different gas pressure. **(a, b)** and **(e, f)** show the output spectral power (SP) for different pre-plasma density ρ_0 . **(c)** and **(g)** show the conversion efficiency. **(d)** and **(h)** show the energy of the output UV pulse. The insets in **(d)** and

(h) show the corresponding UV pulse profile at given conditions, where the propagation distances are taken to be $z = 0.8$ m and $z = 0.92$ m, respectively, where the UV pulse durations are minima.

(Fig. 6d). For the wavelength below 140 nm, the conversion efficiency only changes slightly with pressure, thus the corresponding UV energy is doubled when lowering the pressure. Since the refractive index changes as $n(p) = 1 + p(n_0 - 1)$, where the pressure p is in the unit of atm and $n_0 = n(1 \text{ atm})$, lower pressure can also induce a shorter leaky wavelength for a given pre-plasma density (Fig. 6b). For example, when $\rho_0 = 4 \times 10^{17} \text{ cm}^{-3}$, the leaky wavelength is 160 nm at 1 atm and 142 nm at 0.5 atm.

For helium, both the nonlinear refractive index coefficient and the group velocity dispersion coefficient are much smaller. These characteristics should be beneficial to the generation of high-energy and ultrashort UV pulse. This is further verified by the simulation results of helium (Fig. 6e–h). For helium with 1 atm pressure, the leaky wavelength is too short that the supercontinuum cannot cover it. As a result, in this case we cannot observe isolated spectral hump (not shown). However, we can increase the pressure to achieve a longer leaky wavelength, so that similar spectral hump could be formed (Fig. 6e, f). At $p = 2$ atm, the leaky wavelength could be as short as 91 nm, and the corresponding UV pulse energy is about 25 μJ . For leaky wave in 100–140 nm, the UV pulse energy reaches several hundreds of μJ . Although this requires an input energy ~ 20 mJ, it indeed provides a promising way to generate vacuum UV pulses with rather high energy. For $p = 3$ atm, the UV pulse energy is lower than that when $p = 2$ atm, which is similar to the results of argon. It is noteworthy that the dispersion of helium is much weaker than air and argon in the UV range. For example, the group velocity dispersion coefficients at 200 nm (without plasma) are $3 \text{ fs}^2/\text{cm}$ for air, $1.48 \text{ fs}^2/\text{cm}$ for argon, and $0.05 \text{ fs}^2/\text{cm}$ for helium. As a consequence, the UV pulse with 123 nm wavelength has a shorter pulse duration, only 1.3 fs at the initial stage of leaky wave generation in helium (Fig. 6h). With the increase of the propagation distance in helium-plasma channel, the pulse duration of the UV pulse slightly increases (widened to 2.3 fs after 0.1 m propagation).

For practical uses, people could choose air, argon or helium as the nonlinear medium, depending on the output wavelength required. Besides, the gas pressure could be chosen according to the available laser condition and the required UV pulse energy. For the generation of intense ultrashort UV pulses, helium might be more suitable.

Conclusion

In summary, we have proposed the leaky wave emission model that enables the generation of high-energy and wavelength-tunable ultraviolet pulses. The model was verified by using a preformed plasma channel to modulate

the dispersion condition of gases in simulation. Compared to the resonant dispersive wave scheme in hollow-core fibers, our approach removes the limitation of fiber system, thus showing great potential in energy scaling. The output UV pulse has a tunable wavelength range of 91 to 430 nm and an energy level up to sub-mJ. This study could provide an essential light source for ultrafast science and important driving source for attosecond physics.

Methods

UPPE model

The unidirectional pulse propagation equation (UPPE) that governs the forward-propagating electric field envelope $\hat{E}(k_x, k_y, \omega, z) = \mathcal{F}[E(x, y, t, z)]$ in the pulse local frame ($t \rightarrow t - z/v_g$) is written as²²:

$$\partial_z \hat{E} = i \left(K_z - k_0 - \frac{\omega - \omega_0}{v_g} \right) \hat{E} + \frac{i\omega^2}{2\epsilon_0 c^2 K_z} \mathcal{F} \left[P_{\text{NL}} + \frac{i}{\omega} J \right]. \quad (4)$$

Here $K_z = \sqrt{k(\omega)^2 - k_x^2 - k_y^2}$, $k(\omega) = n(\omega)\omega/c$ is the dispersion relation of the medium, $k_0 = k(\omega_0)$ and $v_g^{-1} = \partial_\omega k(\omega_0)$. The second term on the right side denotes the Fourier transform of time-dependent nonlinear response. The nonlinear polarization includes an instantaneous Kerr response and a delayed Raman contribution²³:

$$P_{\text{NL}} = 2\epsilon_0 n_0 n_2 \left[(1 - \alpha)I + \alpha \int_{-\infty}^t \mathcal{R}(t - t') I(t') dt' \right] E, \quad (5)$$

where the ratio α takes 0.5 for air and 0 for noble gases. The specific form of $\mathcal{R}(t - t')$ that denotes the Raman contribution can be found in the literature¹⁸. The nonlinear refractive index coefficient n_2 takes 0.96×10^{-19} , 1.6×10^{-19} , and $3.6 \times 10^{-21} \text{ cm}^2/\text{W}$ for air, argon, and helium, respectively. The current term is composed of plasma current $J_p = ie^2 \rho E / m_e \omega$ and ionization loss $J_{\text{loss}} = 2W(|E|)U_i \rho_n / E$. The electron density ρ is calculated using the corrected Perelomov-Popov-Terent'ev formula¹⁸:

$$\partial_t \rho = W(|E|) (\rho_n - \rho_0 - \rho), \quad (6)$$

where $W(|E|)$ is the PPT ionization rate, U_i is the ionization potential and ρ_n is the neutral species density. We assume the preformed plasma channel is uniformly distributed with transverse scale much larger than the spot size,

then the effect of initial plasma density ρ_0 can be coupled into the dispersion relation:

$$n(\omega) = n_a(\omega) - \frac{\rho_0}{2\rho_c}. \quad (7)$$

$n_0(\omega)$ is the refractive index of the medium^{24,25}, and $\rho_c = \epsilon_0 m_e \omega^2 / e^2$ is the frequency-dependent critical plasma density that accounts for plasma dispersion.

Ionization model for picosecond CO₂ laser pulses

The ionization rate is composed of three parts:

$$\partial_t \rho = W_{\text{PPT}} \times \rho_{\text{O}_2} + W_{\text{EID}} + W_{\text{Col}} \quad (8)$$

The first term accounts for multi-photon and tunnel ionization of oxygen computed from the PPT model¹⁸. The second term uses a fitting formula to calculate the contribution of EID effect²⁰:

$$W_{\text{EID}} = CE^4 \sqrt{\frac{E^2 + s}{E^2}}, \quad (9)$$

where the fitting parameters are $C = 2.84 \times 10^{-7} \text{ m} \cdot \text{s}^{-1} \text{ V}^{-4}$ and $s = 4.6 \times 10^{18} \text{ V}^2 \text{ m}^{-2}$. The collisional ionization rate for O₂ and N₂ is calculated using the electron temperature dependent model²⁶:

$$W_{\text{Col}} = \sigma_{\text{O}_2} \frac{\rho I}{U_{\text{O}}} + \sigma_{\text{N}_2} \frac{\rho I}{U_{\text{N}}}. \quad (10)$$

$U_{\text{O}} = 12.1 \text{ eV}$ and $U_{\text{N}} = 15.6 \text{ eV}$ are the ionization potential of oxygen and nitrogen, respectively. The collision cross-sections $\sigma_X = e^2 \nu_X(T_e) / m_e \epsilon_0 c \omega_0^2$, where the subscript X indexes O₂, O₂⁺, N₂, and N₂⁺. The collision frequency $\nu_X = 2 \times 10^{-7} \rho_X [\text{cm}^{-3}] (T_e [\text{eV}])^{1/2}$ for molecular species and $\nu_X = 7 \times 10^{-6} \rho_X [\text{cm}^{-3}] (T_e [\text{eV}])^{-3/2}$ for ionic species. The evolution of electron temperature is governed by

$$\partial_t T_e = \frac{2}{3k_B} \sum_X \sigma_X I - \left(\frac{T_e}{U_{\text{O}}} + \frac{2}{3k_B} \right) \sigma_{\text{O}_2} I - \left(\frac{T_e}{U_{\text{N}}} + \frac{2}{3k_B} \right) \sigma_{\text{N}_2} I \quad (11)$$

Spectrogram representation

The spectrogram of the electric field is calculated as follows:

$$P(\tau, \omega) = \left| \int_{-\infty}^{\infty} e^{-i\omega t} E(t) h(t - \tau) dt \right|^2, \quad (12)$$

where the spectrogram $P(\tau, \omega)$ is the function of time delay τ and angular frequency ω . We choose a Gaussian function with 5 fs FWHM as the gate function. The spectrogram is calculated, respectively, for each (x, y) data point, and then summed up.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

L.X. and T.X. discussed and conceived the idea. L.X. performed the theoretical analysis and simulations. L.X. and T.X. analyzed the data and prepared the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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